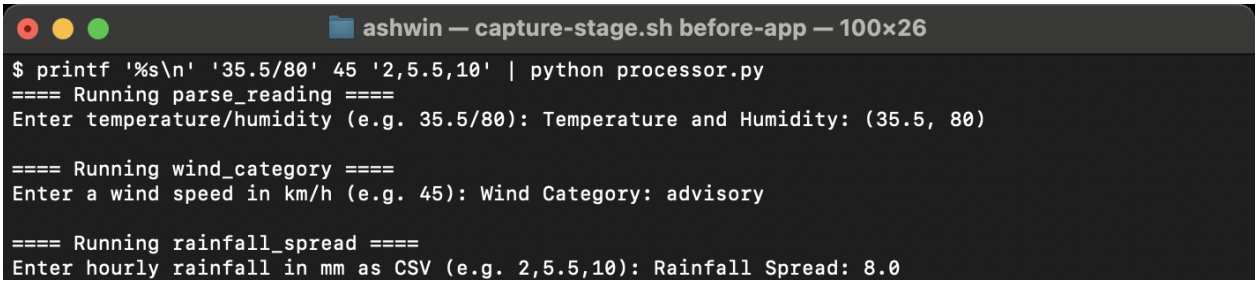
Lab 7: Fuzz Testing

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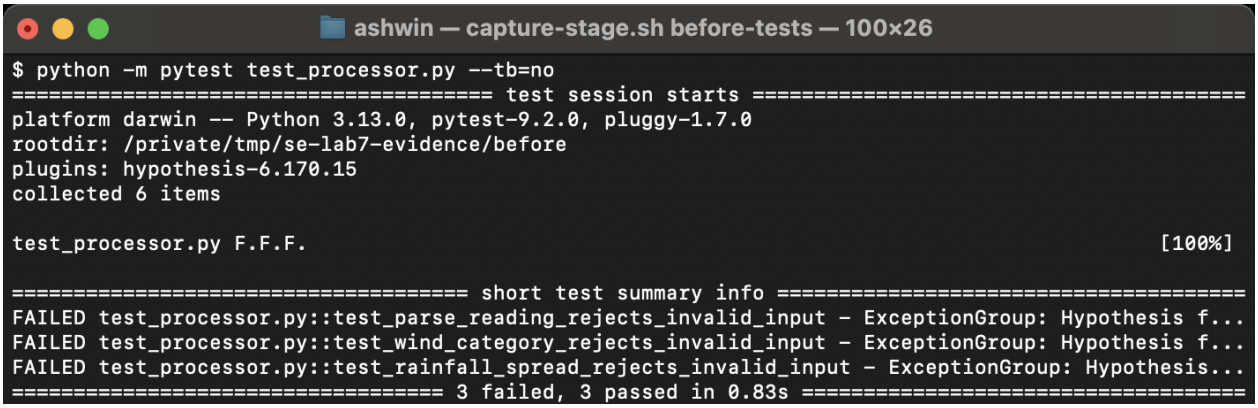
1. Running the starter program

The starter handled the sample readings as expected. It parsed 35.5/80, classified 45 km/h as an advisory and calculated a rainfall spread of 8.0 mm.



2. What the fuzz tests found

For each function, one Hypothesis test checked valid inputs and another checked invalid ones. The three valid-input tests passed. Each invalid-input test found at least one case where the program raised an exception or accepted a bad reading.

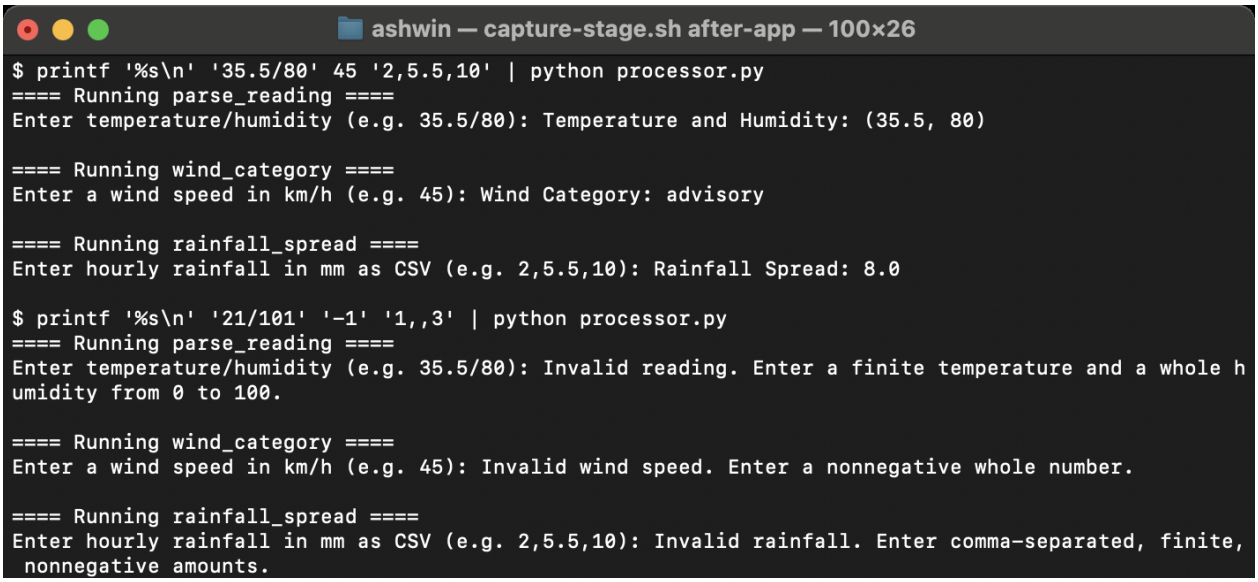


Function	Input	What the starter returned
parse_reading	"nan/50"	A reading with NaN as the temperature.
wind_category	False	The category "none" because False was treated as zero.
rainfall_spread	"0,nan,3"	A spread of 3.0 with a non-finite reading in the list.

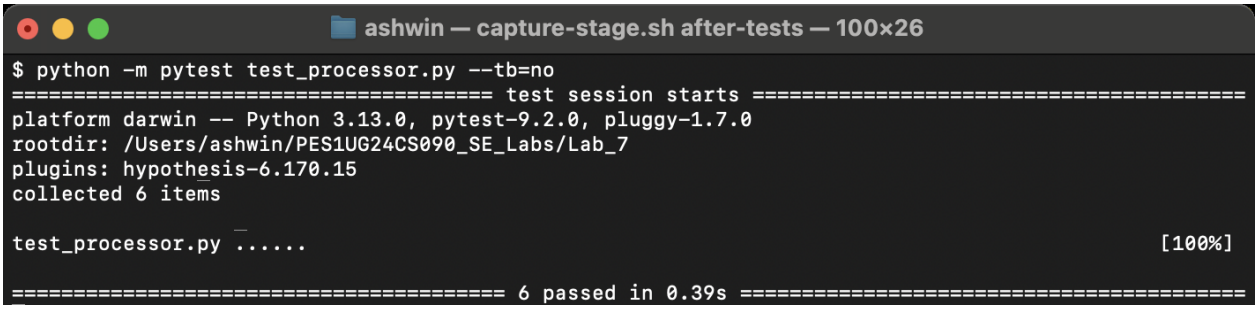
3. Fixing the input checks

The three functions now return None for invalid input. A weather reading needs a finite temperature and a whole humidity percentage from 0 to 100. A wind speed must be a nonnegative whole number. Each rainfall value must be finite and nonnegative, and the list must contain at least one value.

The fixed program still gives the same results for the sample readings. When given 21/101, -1 and 1,,3, it explains what is wrong with each entry.



4. Running the same tests again



All six tests passed after the changes. The valid-input tests also checked the wind thresholds at 39 and 89 km/h and verified that reversing the rainfall readings gives the same spread.

Commands to rerun the tests from Lab_7

```
python -m pip install -r requirements.txt
python -m pytest test_processor.py
```